

Asymmetric Baryogenesis from Electroweak Hysterical Instability in a Cooling Universe

A Standard Model residual mechanism with analytic concurrency at $t \sim 10^{-11}$ s

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

This paper supplies a concrete mechanism for asymmetric baryogenesis in the early Universe from the Hysterical residual of the Standard Model electroweak interaction—not from a new force carrier. Cosmological cooling $T : \infty \rightarrow 0$ through Debye reciprocal stiffnesses with fixed nonzero off-diagonal feedback forces a unique criticality temperature $T_\diamond$: the neutral electroweak residual is subcritical for all $T > T_\diamond$ and supercritical for all $0 < T < T_\diamond$. The same Higgs condensate that shuts off sphalerons defines freeze-out T_* relative to the Debye criticality $T_\diamond$: the concurrent epoch is nonempty if and only if criticality precedes freeze-out ($T_\diamond > T_*$), with no further inequality assumed. A CP-odd roll through that epoch, coupled by $YW\widetilde{W}$ to electroweak topology, produces opposite nonzero baryon yields on the two Hysterical branches: $\eta_B[Y_+] = -\eta_B[Y_-] \neq 0$. That is spontaneous, branch-conditioned asymmetric baryogenesis at the electroweak epoch. Lattice GeV edges convert the concurrent temperatures into a radiation-era clock $t_\diamond \sim 10^{-11}$ s, in agreement with the standard electroweak baryogenesis window. The observed magnitude $\eta_B \sim 10^{-10}$ is not computed here; the mechanism and its analytic concurrency are.

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1 Scope and status

Claim. We provide a mechanism for asymmetric baryogenesis in the early Universe: the Hysterical (off-diagonal) residual of electroweak dynamics, driven supercritical by cosmological cooling, concurrently with active Standard Model $B+L$ violation, selects a broken CP-odd branch and sources a nonzero baryon asymmetry of definite sign on that branch.

Ontology. The response-theory paper defines the stationary Hysterical response of an interaction force observable F by

$$\boxed{F_H = -\text{Tr}\left(F\mathcal{L}^{-1}\mathcal{S}\right)} \quad (1)$$

whenever the inverse is defined on the relevant stable response subspace [1]. No new force carrier is introduced: F_H is a component of the response of F itself. The electroweak interaction therefore already carries the residual whose instability we use. In the reciprocal Hessian of the baryon-asymmetry companion, the coupling g is that off-diagonal feedback, and the partner coordinate is the electroweak condensate / plasma amplitude (this paper).

Mechanism chain.

$$\begin{aligned} &\text{SM EW residual (Debye reciprocal class)} \\ &+ T : \infty \rightarrow 0 \text{ cosmological cooling} \\ \implies &\text{unique } T_\diamond, \text{ high-}T \text{ subcritical;} \\ &\mathcal{C} = (T_*, T_\diamond) \neq \emptyset \iff T_\diamond > T_*; \\ &\text{CP-odd roll on } \mathcal{C} + YW\widetilde{W} \text{ portal} \\ \implies &\text{asymmetric baryogenesis: } \eta_B[Y_+] = -\eta_B[Y_-] \neq 0 \text{ at } t \sim 10^{-11} \text{ s.} \end{aligned} \quad (2)$$

Open phenomenology, not withheld mechanism. What remains open is quantitative: unique SM values of (α_X, α_Y, g) and the precise magnitude $\eta_B \sim 10^{-10}$. Lattice GeV edges of (T_*, T_c) are Standard Model inputs used for the cosmic clock, not outputs of Hysterical Response. None of that retracts the mechanism: cooling forces criticality; the concurrent epoch is characterized exactly by $T_\diamond > T_*$; the portal and branch theorems force a nonzero baryon asymmetry on the selected history whenever that epoch is nonempty and a CP-odd roll occupies it. Late-universe dilution criticality at some gravitational scale factor a_c is a different channel [2]; the present paper is the electroweak baryogenesis mechanism.

Item	Status
Mechanism: EW Hysterical cooling $\Rightarrow$ asymmetric baryogenesis	Claimed (this paper).
Forced unique $T_\diamond$ under Debye cooling	Theorem 3.1 (Lean).
Concurrent epoch $\mathcal{C} = (T_*, T_\diamond)$; nonempty iff $T_\diamond > T_*$	Theorem 4.3 (Lean).
Branch $\eta_B \neq 0$ from concurrent roll	Theorem 5.1.
$t_\diamond \sim 10^{-11}$ s matches EW epoch	Sec. 6.
Lattice GeV edges	SM literature input [5].
Unique (α_X, α_Y, g) / observed η_B	Open phenomenology.

2 Inherited electroweak residual

The baryon-asymmetry companion isolates the two-mode reciprocal quadratic form

$$F_2(X, Y) = \frac{1}{2}k_X X^2 + \frac{1}{2}k_Y Y^2 - gXY, \quad (3)$$

with Hessian determinant $\Delta = k_X k_Y - g^2$, and proves that the neutral point is subcritical, critical, or supercritical according as Δ is positive, zero, or negative [3]. The coupling g is the off-diagonal feedback of the electroweak residual; Y is taken CP-odd and X a CP-even weak-plasma / Higgs-radial partner. Nonzero response does not by itself imply instability: that separation remains in force [3].

Modeling Assumption 2.1 (Debye reciprocal stiffnesses under cooling). On a homogeneous thermal background take

$$\boxed{k_X(T) = \alpha_X T^2, \quad k_Y(T) = \alpha_Y T^2, \quad g = \text{const} \neq 0,} \quad (4)$$

with $\alpha_X, \alpha_Y > 0$. Write

$$\Delta(T) := k_X(T)k_Y(T) - g^2 = \alpha_X \alpha_Y T^4 - g^2. \quad (5)$$

(The optional Higgs restabilization term $\beta_Y m_W(T)^2$ is deferred to Corollary 3.3.)

Equation (4) packages Debye-like plasma stiffnesses of the *same* electroweak sector whose topology supplies $B+L$ violation. Cooling is driven by cosmological expansion: we assume a continuous strictly decreasing temperature history with

$$\lim_{t \rightarrow t_{\text{hot}}} T(t) = \infty, \quad \lim_{t \rightarrow t_{\text{cold}}} T(t) = 0 \quad (6)$$

on an interval of cosmic time (physical input: radiation-era FLRW thermal history).

3 Forced cooling criticality

Theorem 3.1 (Forced reciprocal criticality under cooling). *Fix $\alpha_X, \alpha_Y > 0$ and $g \neq 0$ in Model 2.1. Define*

$$T_\diamond := \left(\frac{g^2}{\alpha_X \alpha_Y} \right)^{1/4} > 0. \quad (7)$$

Then:

- (i) $\Delta(T_\diamond) = 0$ (critical);
- (ii) $\Delta(T) > 0$ (subcritical) for all $T > T_\diamond$;
- (iii) $\Delta(T) < 0$ (supercritical) for all $0 < T < T_\diamond$;
- (iv) $T_\diamond$ is the unique positive root of $\Delta(T) = 0$.

In particular, every continuous cooling history (6) that begins at sufficiently high temperature is necessarily subcritical and crosses criticality exactly once, at the unique temperature $T_\diamond$.

Proof. Substitute (7) into (5): $\Delta(T) = \alpha_X \alpha_Y (T^4 - T_\diamond^4)$. The sign of $\Delta(T)$ for $T > 0$ is the sign of $T - T_\diamond$, and $\Delta(T_\diamond) = 0$. Uniqueness of the positive root is immediate from strict monotonicity of $T \mapsto T^4$ on $(0, \infty)$. $\square$

Remark 3.2 (Converse to open-coefficient placement). For every target temperature in an open interval one can *choose* g to place Debye criticality there (dual of Theorem 3.1; Lean: open-placement package). Theorem 3.1 is the converse direction: for any fixed nonzero off-diagonal coupling, cooling from infinity *forces* a unique crossing.

Corollary 3.3 (Optional Higgs restabilization). *If one restores $k_Y(T) = \alpha_Y T^2 + \beta_Y m_W(T)^2$ with $\beta_Y > 0$ and a mean-field m_W that vanishes above a condensate scale T_c and grows below it, then Δ may return through zero at some $T_{\text{re}} < T_\diamond$. Existence of such a bounded supercritical band is an open condition on $(\alpha_X, \alpha_Y, \beta_Y, g)$ (Lean: Higgs-restabilization witness). Above T_c the extra term vanishes, so Theorem 3.1 continues to control the symmetric-phase crossing.*

4 Condensate-coupled activity and analytic overlap

Electroweak topology and the Hysterical residual belong to the same interaction. The Higgs condensate scale that defines the electroweak crossover is therefore *not* an external bolt-on to the reciprocal Hessian: it is the same thermal order parameter that (i) truncates Debye stiffnesses above the crossover and (ii) raises the sphaleron barrier below it.

Assumption 4.1 (Shared condensate scale). There is a crossover temperature $T_c > 0$ such that the electroweak condensate (and hence m_W) vanishes for all $T \geq T_c$ and is nonzero for $T < T_c$. There is a freeze-out temperature $0 < T_* < T_c$ below which the sphaleron rate falls below the Hubble expansion. Standard Model electroweak topology is active for every $T > T_*$; in particular it is active throughout the symmetric phase $T \geq T_c$ [5].

Assumption 4.1 is Standard Model thermal dynamics. It does *not* insert the lattice GeV numbers; those appear only in Sec. 6 for a numerical clock.

Definition 4.2 (Concurrent residual-topology set). Under Model 2.1 and Assumption 4.1, write

$$\mathcal{C} := \{T > 0 : \Delta(T) < 0 \text{ and } T > T_*\}. \quad (8)$$

Elements of $\mathcal{C}$ are temperatures at which the electroweak residual is linearly supercritical while electroweak topology is still active.

Theorem 4.3 (Concurrent epoch: sharp characterization). *Assume Model 2.1 and Assumption 4.1, and let $T_\diamond$ be as in Theorem 3.1. Then for every $T > 0$,*

$$\boxed{T \in \mathcal{C} \iff T_* < T < T_\diamond.} \quad (9)$$

Equivalently $\mathcal{C} = (T_, T_\diamond)$ as an open interval (empty if $T_\diamond \leq T_*$). In particular*

$$\boxed{\mathcal{C} \neq \emptyset \iff T_\diamond > T_*} \quad (10)$$

No independent inequality on (α_X, α_Y, g) is assumed: nonempty concurrency is exactly the statement that Debye criticality precedes sphaleron freeze-out.

Proof. By Theorem 3.1, for $T > 0$ one has $\Delta(T) < 0$ if and only if $T < T_\diamond$. By Assumption 4.1, topological activity holds if and only if $T > T_*$. Intersecting these conditions yields (9), and taking the interval nonempty yields (10). $\square$

Remark 4.4 (Symmetric-phase special case). If in addition $T_\diamond > T_c$, then $(T_c, T_\diamond) \subseteq \mathcal{C}$ and concurrency occupies the whole remaining symmetric phase above the crossover. That is a special case of (10), not an extra hypothesis required for the mechanism.

Corollary 4.5 (Criticality before freeze-out). *For fixed $\alpha_X, \alpha_Y > 0$ and $T_* > 0$, one has $T_\diamond > T_*$ if and only if $|g| > \sqrt{\alpha_X \alpha_Y} T_*^2$.*

5 Asymmetric baryogenesis from the concurrent epoch

Whenever $T_\diamond > T_*$, Theorem 4.3 supplies a nonempty concurrent interval $(T_*, T_\diamond)$ —the arena in which baryogenesis occurs. A physically realized CP-odd electroweak residual Y couples to electroweak topology by

$$\mathcal{L}_{\text{int}} = \frac{c_H}{f_H} Y W_{\mu\nu}^a \widetilde{W}^{a\mu\nu}, \quad \frac{c_H}{f_H} \neq 0, \quad (11)$$

as in the portal construction [3, 4]. A rolling history induces a Chern–Simons chemical potential proportional to $\dot{Y}$; detailed balance biases topological rates; the anomaly converts that bias into a baryon source odd in Y . Spontaneous selection of a supercritical branch $Y_\pm$ therefore produces a baryon excess of definite sign on that history—asymmetric baryogenesis, with the opposite sign on the opposite branch.

Theorem 5.1 (Electroweak Hysterical baryogenesis). *Assume Model 2.1, Assumption 4.1, the portal (11), and $T_\diamond > T_*$ (equivalently $\mathcal{C} \neq \emptyset$). Let $Y_\pm$ be opposite CP-odd histories whose washout-weighted source has nonzero support on $\mathcal{C} = (T_*, T_\diamond)$, on a common even thermal background with $\eta_B(t_i) = 0$. Then*

$$\boxed{\eta_B[Y_+] = -\eta_B[Y_-] \neq 0.} \quad (12)$$

This is the baryogenesis mechanism of the present paper: cooling-driven electroweak residual instability concurrent with active SM $B+L$ violation, converted by (11) into a branch-conditioned baryon asymmetry [3].

Proof. Theorem 4.3 gives $\mathcal{C} = (T_*, T_\diamond) \neq \emptyset$. A roll with nonzero overlap against topology on that epoch is the concurrency package for the $W\bar{W}$ instance of the master branch-asymmetry theorem [3]. Those theorems supply (12). $\square$

No preferred global sign is required in the underlying equations: labelling of Y_+ versus Y_- is arbitrary [3]. Our Universe realizes one branch; the mechanism explains why that branch carries $\eta_B \neq 0$.

6 Cosmic time of the mechanism

The concurrent temperatures are already fixed by Theorem 4.3 once $T_\diamond > T_*$. Inserting Standard Model lattice edges converts them into a cosmic clock and places the mechanism in the standard electroweak baryogenesis window.

Assumption 6.1 (Radiation-era temperature–time law). In a radiation-dominated FLRW universe with approximately constant effective relativistic degrees of freedom g_* ,

$$t \simeq \frac{2.42}{\sqrt{g_*}} \left(\frac{\text{MeV}}{T} \right)^2 \text{ s} \quad (13)$$

[6].

Assumption 6.2 (Lattice electroweak edges). On the lattice [5],

$$T_c \simeq 159.5 \text{ GeV}, \quad T_* \simeq 131.7 \text{ GeV}. \quad (14)$$

Corollary 6.3 (Baryogenesis at $t \sim 10^{-11} \text{ s}$). *Under Assumptions 6.1–6.2 and $T_\diamond > T_*$, the mechanism of Theorem 5.1 operates on $\mathcal{C} = (T_*, T_\diamond)$ at electroweak temperatures, hence at*

$$t_\diamond := t(T_\diamond) \sim 10^{-11} \text{ s} \quad (15)$$

whenever $T_\diamond$ is within a few factors of the electroweak scale. Explicitly, with $g_* \sim \mathcal{O}(10^2)$ and $T \sim 150 \text{ GeV} = 1.5 \times 10^5 \text{ MeV}$,

$$t \sim \frac{2.42}{10} (1.5 \times 10^5)^{-2} \text{ s} \sim 10^{-11} \text{ s}.$$

This matches the standard cosmic age of the electroweak epoch ($\sim 10 \text{ ps}$), as required for an electroweak-scale baryogenesis mechanism.

7 Logical status of the results

Quantity / claim	Status
Asymmetric baryogenesis mechanism (EW Hysterical cooling)	Claimed.
F_H residual of the electroweak interaction	Definitional [1].
Forced unique $T_\diamond$ under Debye cooling	Theorem 3.1 (Lean).
Higgs restabilization band	Corollary 3.3.
Shared condensate; activity for $T > T_*$	Assumption 4.1 (SM).
$\mathcal{C} = (T_*, T_\diamond)$; nonempty iff $T_\diamond > T_*$	Theorem 4.3 (Lean).
$T_\diamond > T_* \Leftrightarrow g > \sqrt{\alpha_X \alpha_Y} T_*^2$	Corollary 4.5.
Branch $\eta_B \neq 0$ (baryogenesis on the selected history)	Theorem 5.1.
$t_\diamond \sim 10^{-11} \text{ s}$ in the EW window	Corollary 6.3.
Lattice GeV edges	SM literature [5].
Extra inequality $T_\diamond > T_c$	Not required (Remark 4.4).
Unique SM (α_X, α_Y, g)	Open phenomenology.
Observed magnitude $ \eta_B \sim 10^{-10}$	Open phenomenology.

8 Conclusion

Asymmetric baryogenesis in the early Universe is realized here as a property of the Standard Model electroweak interaction under cosmological cooling. The electroweak Hysterical residual—off-diagonal feedback in the response of that same force—is forced through a unique criticality $T_\diamond$ as $T : \infty \rightarrow 0$. Sharing the SM freeze-out scale T_* with the Debye Hessian gives the sharp concurrency law $\mathcal{C} = (T_*, T_\diamond) \neq \emptyset$ if and only if $T_\diamond > T_*$: criticality before freeze-out, with no further inequality assumed. A CP-odd roll through that epoch sources

$$\eta_B[Y_+] = -\eta_B[Y_-] \neq 0 \tag{16}$$

at $t_\diamond \sim 10^{-11}$ s, the standard electroweak cosmic time.

What remains for phenomenology is the magnitude of η_B and the precise microscopic values of the Debye coefficients—not the existence of the mechanism, nor an ad hoc placement of $T_\diamond$ above T_c .

References

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